

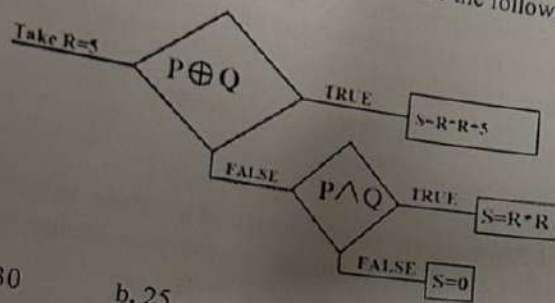
For the second part; please explain your answers step by step.

Good luck!

Part I. Multiple Choice Questions (q / 2 pts)

QUESTIONS

Q1. Let $P: 5+10=15$, $Q: 5*10=50$ and consider the following flow chart of a computer program. Then the value of S is:



- a. 30 b. 25 c. 0 d. 10 e. 15

Q2. Suppose $f: \mathbb{Z} \rightarrow \mathbb{R}$ has the rule $f(n) = 2n - 3$. Which of the following is the range of f ?

- a. the set of natural numbers $\{0, 1, 2, 3, \dots\}$.
 b. the set of integers $\{\dots, -2, -1, 0, 1, 2, 3, \dots\}$.
 c. the set of odd integers.
 d. the set of even integers.
 e. the real numbers.

Q3. Suppose f and g are functions from $\mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = \left\lfloor \frac{x}{2} \right\rfloor$ and $g(x) = \left\lfloor \frac{5-2x}{3} \right\rfloor$. Find $(f \circ g)(5)$.

- a. 1. b. -1. c. 0. d. None of these e. 2

Q4. In Inclusion-Exclusion Principle, if A and B are any two finite sets then _____?

- a. $n(A \cap B) = n(A) + n(B) - n(A \cap B)$
 b. $n(A \cup B) = n(A) + n(B) - n(A \cup B)$
 c. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
 d. $n(A \cup B) = n(A) + n(B) + n(A \cap B)$
 e. $n(A \cap B) = n(A) + n(B) + n(A \cap B)$



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Q5. Which one of these rules defines a function f from the set of all strings of length six of letters of the alphabet to the set $\{1, 2, 3, 4, 5, 6\}$?

- a. The rule that assigns to each string the number of vowels in the string. For example, $f(\text{TAZNAV}) = 2$.
- b. The rule that assigns to each string the number of times Z is an element of the string. For example, $f(\text{RZVZQC}) = 2$.
- c. The rule that assigns to each string the reverse of the string. For example, $f(\text{BAQKDU}) = \text{UDKQAB}$.
- d. The rule that assigns to each string the position in which the first Z occurs. For example, $f(\text{PPABZY}) = 5$.
- e. The rule that assigns to each string the number of distinct letters appearing in the string. For example, $f(\text{TNVRRN}) = 4$.

Q6. There are 15 mathematics majors and 320 computer science majors at a college. In how many ways can two representatives be picked so that one is a mathematics major and the other is a computer science major? (Note: There is no one who is a major in both fields.)

- a. 335
- b. 345
- c. 305
- d. 4850
- e. 4800

Q7. There are 15 mathematics majors and 320 computer science majors at a college. In how many ways can one representative be picked who is either a mathematics major or a computer science major? (Note: There is no one who is a major in both fields.)

- a. 335
- b. 345
- c. 4800
- d. 4850
- e. 5000

Q8. You have 12 balls, numbered 1 through 12, which you want to place into 4 boxes, numbered 1 through 4. If boxes can remain empty, in how many ways can the 12 balls be distributed among the 4 boxes?

- a. $\frac{12!}{4!}$
- b. 12^4
- c. $C(12,4)$
- d. $P(12,4)$
- e. 4^{12}

Q9. How many bit strings (strings of 0's and 1's) are there of length 7 that have more 0's than 1's?

- a. $\frac{2^7}{2}$
- b. $2^4 + 2^5 + 2^6 + 2^7$
- c. $C(7,4) + C(7,5) + C(7,6) + C(7,7)$
- d. $C(7,4) \cdot C(7,5) \cdot C(7,6) \cdot C(7,7)$
- e. $2 \cdot 2 \cdot 2 \cdot 1 \cdot 1 \cdot 1 \cdot 1$



Q10. How many different license plates are available if the license plate pattern consists of four letters followed by three digits? (Assume that all letters are uppercase and the digits are 0, 1, ..., 9.)

- a. $26 \cdot 25 \cdot 24 \cdot 23 \cdot 10 \cdot 9 \cdot 8$
- b. $26^4 \cdot 10 \cdot 9 \cdot 8$
- c. $26^4 + 10^3$
- d. $26^4 \cdot 10^3$
- e. $C(26,4) \cdot C(10,3)$

Q11. How many strings of all the first seven letters of the alphabet (A, B, C, D, E, F, G) are there that contain no repeated letters and begin or end with a vowel (A or E)?

- a. $7! - 5!$
- b. $2 \cdot 6! + 2 \cdot 6!$
- c. $6! + 6!$
- d. $6! + 6! - 5!$
- e. $2 \cdot 6! + 2 \cdot 6! - 2 \cdot 5!$

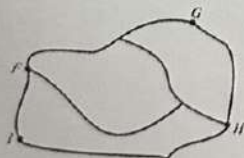
Q12. How many different passwords are possible if each password consists of six characters where each character is either an uppercase letter, a lowercase letter, or a digit, and at least one digit must be included in the password? (There are 26 possible uppercase letters, 26 possible lowercase letters, and 10 possible digits.)

- a. $62^6 - 10^6$
- b. $62^6 - 26^6$
- c. $62^6 - 36^6$
- d. $62^6 - 52^6$
- e. $62^6 - 62^5$

Q13. How many terms are there in the expansion of $(x + y)^{10}$?

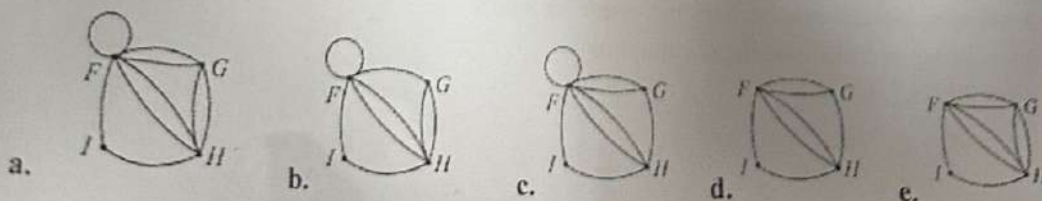
- a. 9
- b. 10
- c. 11
- d. 12
- e. 13

Q14. The diagram shows a map of the roads between four towns, F, G, H and I.



on the map is:

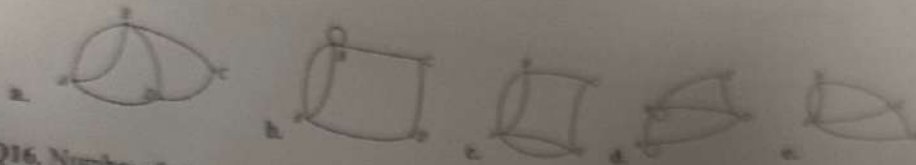
A network diagram that represents all the connections between the four towns





Q15. A graph corresponding to the adjacency matrix shown below is:

	A	B	C	D
A	0	1	0	1
B	1	0	1	1
C	0	1	0	1
D	1	1	1	0



Q16. Number of edges incident with the vertex v is called ____?

- a. Handshaking lemma
- b. Degree of an edge
- c. None of the above
- d. Degree of a graph
- e. Degree of vertex

Q17. Which of the following is not a type of graph?

- a. Bipartite graph
- b. Complete graph
- c. Isomorphic graph
- d. Scalar graph
- e. Pseudograph

Q18. When two graphs such as $G(V, E)$ and $G^*(V^*, E^*)$ have a one-to-one correspondence, they are said to be ____?

- a. Isomorphic
- b. Complete
- c. Planar
- d. Regular
- e. Bipartite

Q19. What is a graph in graph theory?

- a. A bar chart used to represent data
- b. A mathematical structure composed of vertices/nodes connected by edges
- c. A diagram showing relationships between data points
- d. A visual representation of statistical trends
- e. A graph of a function



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Q30. Use notation to rewrite the sum: (2 pts)
 $1 + 2 + 4 + 8 + \dots + 250$. Answer:

Q31. Consider the sequence given by $a_n = 6.5n - 1$.
i. Find the first 4 terms of the sequence. (4 pts)

$a_1 =$ _____

$a_2 =$ _____

$a_3 =$ _____

$a_4 =$ _____

ii. What sort of sequence is this? (1 point)

☐ arithmetic

☐ geometric

☐ neither

Q32. Give a formula for the coefficient of x^k in the expansion of $(x^2 - \frac{1}{x})^{100}$, where k is an integer. (8 pts)

Answer:

Q33. A graph has 24 vertices and 30 edges. It has five vertices of degree 4, seven pendant vertices, and seven vertices of degree 2. All other vertices have degree 3 or 4. How many vertices of degree 4 are there? (4 pts)

Answer:

Q34. Suppose the data elements are 22, 8, 14, 17, 3, 9, 27, 11, 35. Build a binary search tree for these data elements. (4 pts)

Answer:

Q35. Can a simple graph exist with 15 vertices each of degree five? Explain briefly. (3 pts)

Answer:

Q36. Suppose that there are four employees in the computer support group of the School of Engineering of a university. Each employee will be assigned to support one of four different areas: hardware, software, networking, and wireless. Suppose that A is qualified to support hardware, networking, and wireless; B is qualified to support software and networking; C is qualified to support networking and wireless, and D is qualified to support hardware and software.



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Q20. In a strongly connected directed graph:

- a. Every vertex has the same degree
- b. There is a path from every vertex to every other vertex
- c. There are no cycles
- d. The graph has no directed edges
- e. None of the above

Q21. In a directed graph, the sum of in-degrees and out-degrees of all vertices is:

- a. Equal to the number of edges
- b. Equal to the number of vertices
- c. Twice the number of edges
- d. Twice the number of vertices
- e. None of the above

Q22. A connected planar graph having 6 vertices, 7 edges contains _____ regions.

- a. 11
- b. 9
- c. 7
- d. 5
- e. 3

Q23. A Hamiltonian path in a graph is:

- a. A path that visits every vertex exactly once
- b. A path that starts and ends at the same vertex
- c. A path that covers all edges exactly once
- d. A path that contains the maximum number of edges
- e. None of the above

Q24. The length of a path in a graph is:

- a. The number of regions
- b. The number of vertices
- c. The sum of the degrees of vertices
- d. The number of connected components
- e. The number of edges